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Seyed Moeen Nehzati

Education

2022–2027 **PhD in Economics**.
New York University, NY, USA

2015–2019 **BS in Computer Engineering**.
Sharif University of Technology, Tehran, Iran
cumulative GPA: **19.08/20**

Research Interest

Optimization - Optimal Transport - Neural Networks - Experimental Design.

Work Experience

2020-2022 **Predoctoral research fellow**, *UCLA and USC*.
I worked on disentangling the complex relationship between genetic factors and behavioral traits. I was responsible for solving statistical models and implementing efficient solutions scalable to terabytes of data. This effort led to papers published in Nature and the development of statistical software regularly used by practitioners in the field.

2019 **Data Scientist**, *Cafe bazaar*.
Cafe Bazaar is an Iranian Android marketplace with more than 50 million active users. There, I worked on optimizing the advertising auctions and evaluating them empirically. Due to my technical background, I also participated in the creation, implementation, and deployment of the machine learning models needed for resolving auctions.

Research

2025 **Universal Representation of Generalized Convex Functions and their Gradients**, *arXiv*.

2025 **Family-based genome-wide association study designs for increased power and robustness**, *Nature Genetics*.

2024 **Family-GWAS reveals effects of environment and mating on genetic associations**, medRxiv.

2022 **Mendelian imputation of parental genotypes improves estimates of direct genetic effects**, *Nature Genetics*.

2022 **Polygenic prediction of educational attainment within and between families from genome-wide association analyses in 3 million individuals**, *Nature Genetics*.

2020 **Building stable off-chain payment networks**, arXiv.

Software

gconvex, *available here* .
Implements Finitely Convex Models developed in Universal Representation of Generalized Convex Functions and their Gradients. The models are used to find optimal selling mechanism with single buyer and multiple goods. The results match the theoretical optimal benchmark and are quick to be found even on personal computers.

SNIPar, *available here* .
A python library for fast imputation of missing parental genotypes from observed genotypes in a nuclear family, performing family-based genome-wide association and polygenic score analysis

poissonreg, *available here* .
A lightweight python package for fast sparse poison regression based on pytorch

Technical skills

Python/Java/Cython Expert
Scientific Programming Expert
Pytorch/Scikit/Spark Experienced
C/C++/R/SQL/Julia Experienced

Awards and Honors

Among top 0.05% in Iran's national university entrance exam.
Member of Iran's National Elites Foundation.

Languages

Persian(native)/ English(Fluent)/ Arabic(basic)

References

name	email address
◦ Prof. Alfred Galichon Courant Institute of Mathematical Sciences & Department of Economics, NYU	◦ ag133@nyu.edu
◦ Prof. David Cesarini Department of Economics, NYU	◦ dac12@nyu.edu
◦ Prof. Daniel Benjamin Anderson School of Management, UCLA	◦ daniel.benjamin@anderson.ucla.edu